Hydrogen Bonds

Hydrogen bonds are weak electrostatic interactions between an electro negative atom and a hydrogen which is covalently linked to another electro negative atom.

α Helix

*I*t is a spiral (helical) structure of a tightly packed and coiled main chain of a polypeptide with the side chain groups of amino acids protruding outside. The helical structure is achieved by the formation of hydrogen bonds between the –C=O of an nth amino acid with the –NH group of $n+4^{th}$ amino acid. Each turn of an α helix contains 3.6 amino acids.The α helices are mostly right handed but there are rare instances where left handed α helices are also present in proteins. The amino acid Proline can produce a kink in an α helix as its secondary amino group is not geometrically compatible inside an α helix.

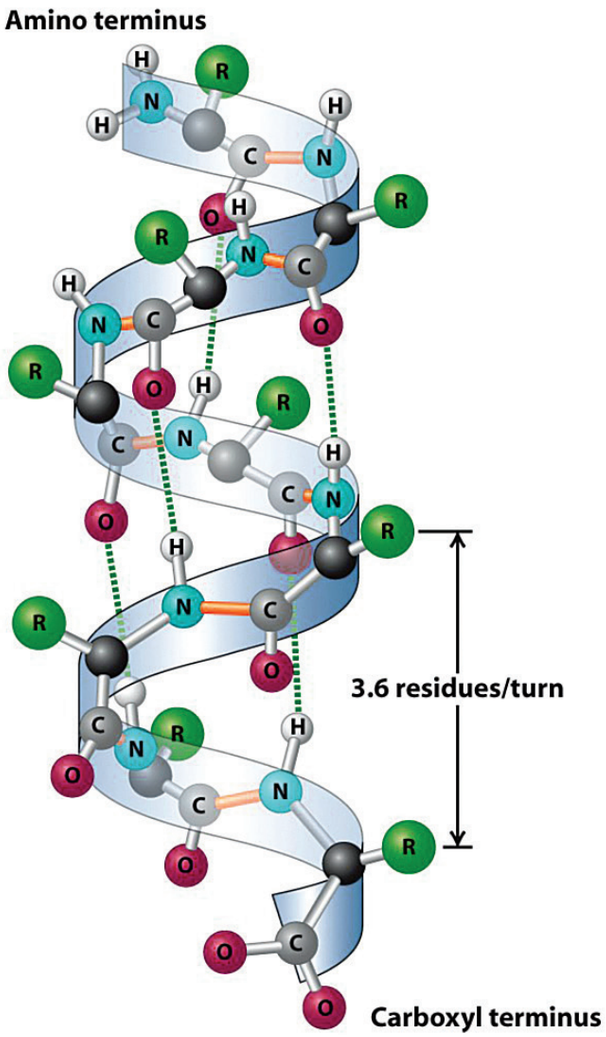


Figure 3.9 Structure of an α helix

Unit 3.indd 87 21-05-2018 15:09:32